

Strategic Analysis Report: Hotel Booking Demand & Revenue Optimization

Date: March 2026

Tools Used: SQL (SQLite), VS Code, DB Browser

Dataset: Kaggle Hotel Booking Demand (119,390 records)

1. Executive Summary

This report provides a multi-dimensional diagnostic of the hotel group's operational health. Our analysis reveals a significant **bifurcation** between City and Resort hotel performance. While **City Hotels** drive high volume and premium last-minute pricing, they suffer from a **critical 42% cancellation rate**.

Conversely, **Resort Hotels** maintain higher stability but struggle with “perishable inventory” price slashing for late bookings.

Key Strategic Pivot: Shift focus from “Booking Volume” to “Booking Quality” through **risk-based cancellation controls for long-lead bookings** and targeted acquisition of high-yield Western European transient markets.

2. Phase 1: Data Integrity & Sanitization

Before analysis, the dataset was sanitized to ensure business logic consistency:

Ghost Bookings: Removed records with 0 adults, children, and babies.

Price Anomalies: Filtered out negative ADR (Average Daily Rate) values.

Missing Values: Categorized undefined countries as “Unknown” and null agents/companies as “0”.

Final cleaned dataset: ~ 117,399 valid records.

3. Phase 2: Global Overview (The Landscape)

Comparing the two properties reveals a fundamental difference in risk profiles:

Hotel	Total bookings	Cancel rate	Avg_adr	Avg_lead_time	Avg_stay_nights
City Hotel	78,092	42.22	106.95	111.0	3.0
Resort Hotel	39,307	28.08	96.77	93.4	4.37

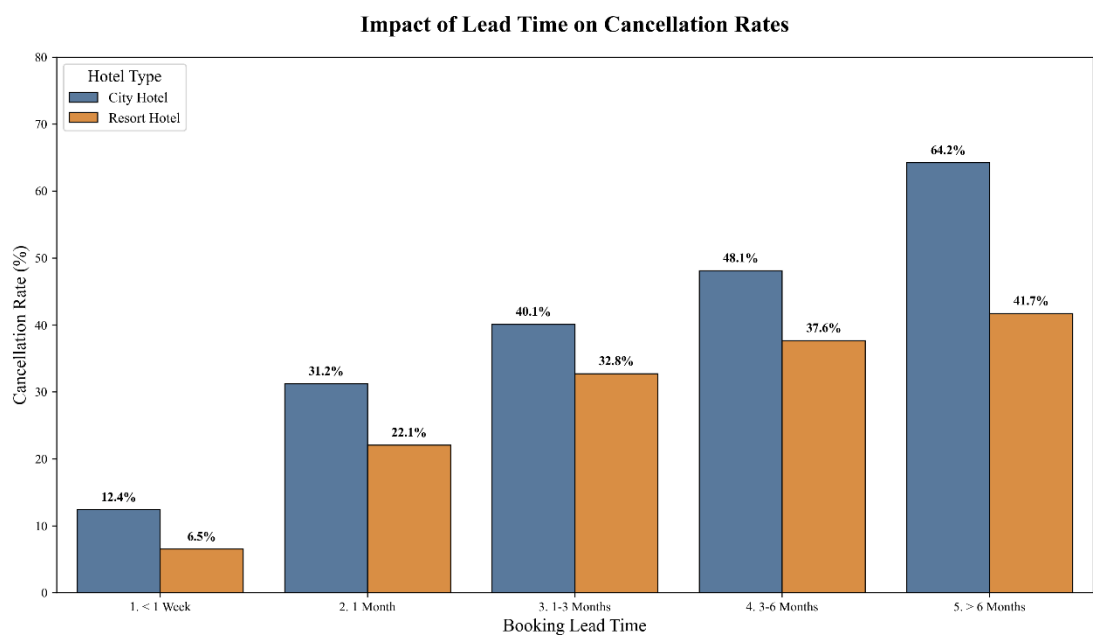
Compared with Resort Hotel, City Hotel has a significantly higher cancel rate despite its higher number of total bookings.

4. Phase 3: Loss Diagnosis (The Cancellation Crisis)

4.1 The “Death Zone” of Lead Time

We discovered that lead time is the strongest predictor of cancellation. For City Hotels, bookings made more than **180 days** in advance have a **64.2% failure rate**.

Conversely, booking made within 7 days of arrival have only a **12%** cancellation rate.



4.2 The Deposit Fallacy

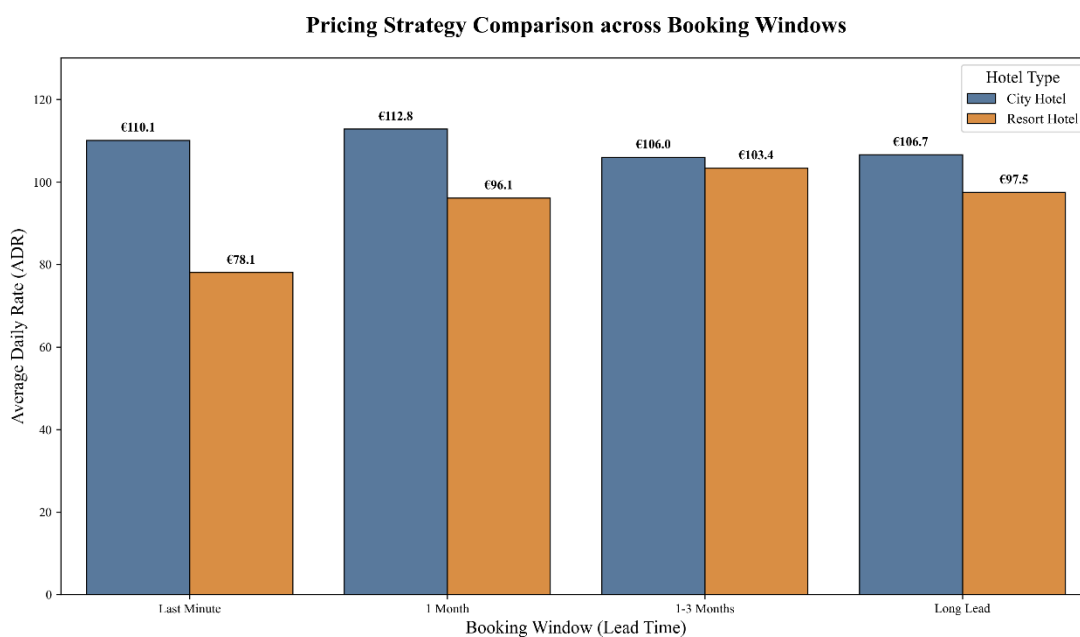
Contrary to intuition, **Non-Refundable** bookings in our data showed a **99.8% cancellation rate**. These are almost exclusively linked to “Group” segments and “Offline Travel Agents.” This suggests a systemic failure in the B2B channel where entire blocks are reserved and subsequently defaulted on.

5. Phase 4: Revenue Mining (The Profit Engine)

5.1 Pricing Strategies: Harvest vs. Clearance

City Hotels (Harvesting): Effectively utilize a “Last-Minute Premium” strategy. ADR for bookings within 1 week is **€110**, compared to €106 for long-lead bookings.

Resort Hotels (Clearance): Suffer from “Perishable Inventory” panic. Last-minute ADR drops to **€78**, significantly lower than the €103 peak for 1–3-month bookings.



5.2 The “Family” Premium

City Hotel: Family ADR is **€149** vs. Adults-only €104.

Resort Hotel: Family ADR is **€154** vs. Adults-only €86.

6. Phase 5: Customer Personas & Loyalty

6.1 The Anchor of Stability: Repeated Guests

Repeated guests are the foundation of operational security. In Resort Hotels, their cancellation rate is a negligible **6.7%**, compared to 29% for new guests.

6.2 The “Contract” Segment Failure

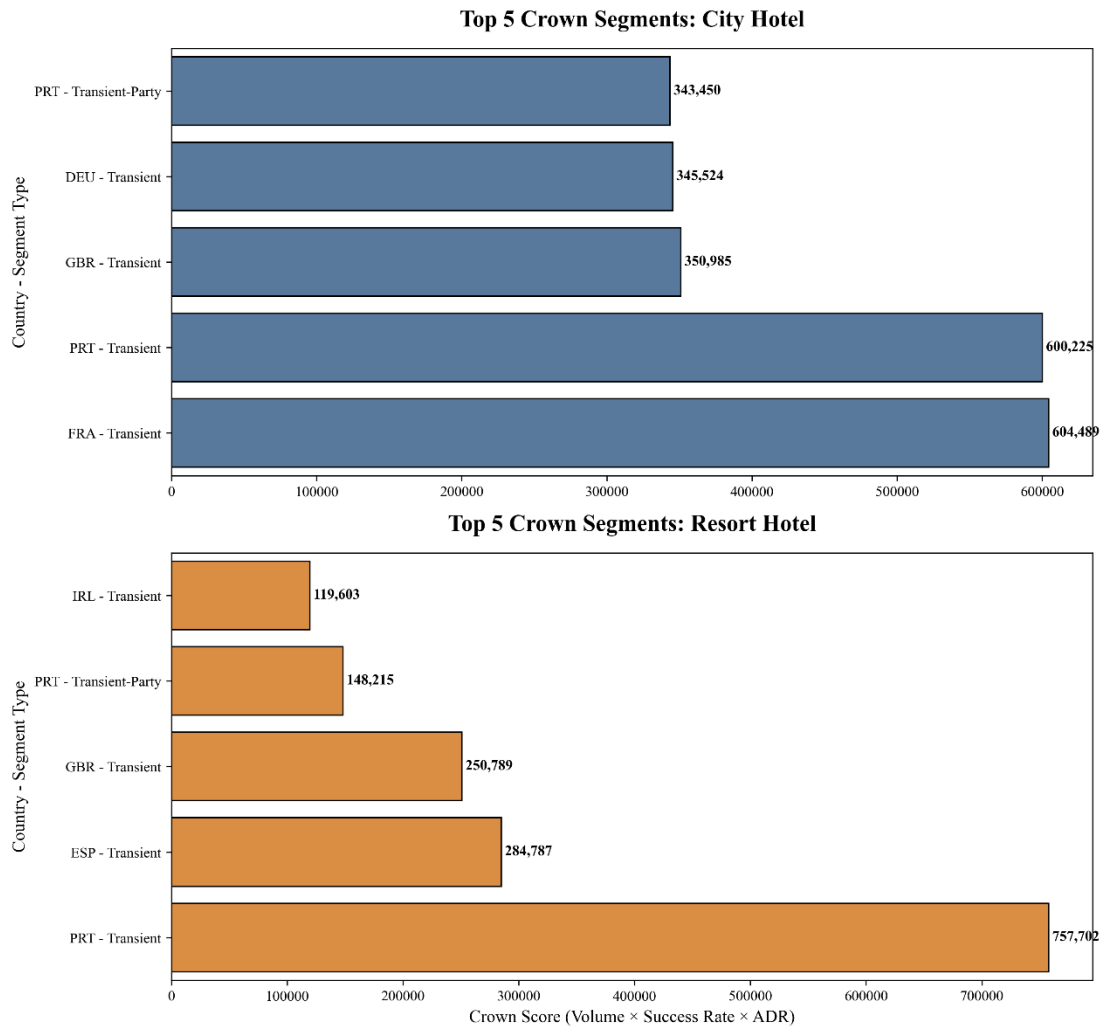
In City Hotels, **Contract** bookings (corporate agreements) are performing poorly with a **48% cancellation rate**, proving less reliable than standard transient (walk-in) customers.

7. Phase 6: The Crown Jewel Analysis

To identify our most valuable markets, we developed a **Crown Score**:

$$Score = Total\ Volume \times Success\ Rate \times ADR$$

Hotel	Country	Customer Type	Total Bookings	Success Rate	Average ADR	Crown Score
City Hotel	FRA	Transient	6,978	75.95%	114.05	604,489.01
	PRT	Transient	20,194	28.87%	102.95	600,225.42
	GBR	Transient	4,636	66.98%	113.04	350,985.65
Resort Hotel	PRT	Transient	13,698	57.02%	97.02	757,702.72
	ESP	Transient	3,222	74.27%	119.01	284,787.41
	GBR	Transient	3,841	77.84%	83.88	250,789.04



8. Final Strategic Recommendations

8.1 City Hotel

Adopt Risk-Based Cancellation Controls: Apply stricter cancellation controls to long-lead bookings, differentiated by customer segment and booking channel rather than using a blanket non-refundable policy.

Review High-Risk B2B Channels: Investigate and renegotiate Group and Offline Travel Agent arrangements associated with unusually high cancellation rates before expanding non-refundable policies.

Capitalize on the “French Gold”: Allocate a significant share of the digital marketing budget toward French transient travelers, who show the best balance of price and reliability.

8.2 Resort Hotel

Implement Last-Minute Pricing: Stop aggressive price slashing for late bookings. Instead, offer “Value-Add” (free spa, dinner) to maintain ADR while filling rooms.

Cultivate the Family Premium Segment: Since Families pay a €60+ premium, invest in “Kids-Club” amenities and family-centric marketing to increase the volume of this high-margin segment.

The “British Anchor”: Deepen relationships with GBR (UK) agents. Their 99.9% success rate in the Contract segment makes them the perfect “occupancy filler” for low seasons.